A $10 Sensor Uses Light to Test Things Like a Lab

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Scientists at the University of Cambridge in England have built a tiny light **sensor** that costs about $10. It can measure things as well as big lab machines.

The **sensor** is a **spectrometer**. It studies light that touches an object. Different materials change light in different ways, so the light shows what something is made of.

Small **spectrometers** are usually not very accurate. The Cambridge team used a clever math idea to make theirs work better. It is just a few **centimeters** wide and works in less than a second.

In tests, it correctly named plastics, medicines, coffee, flour and tea every time. It also measured things in the body, like skin **moisture** and blood sugar, in lab conditions.

One day, it might help people check blood sugar without needles. But the scientists say more testing is needed first. The **sensor** also worked in very cold and very hot temperatures.

Word Treasure

sensor -- A device that notices or measures something.

spectrometer -- A tool that studies light to find out what something is made of.

centimeters -- Small units for measuring length; about the width of a fingernail each.

moisture -- A small amount of wetness.

Background you need

Every material changes light in its own special way, a bit like a fingerprint.

Scientists use tools that read light to find out what things are made of, but the best ones are usually big and costly.

Break it down

WHO: Scientists at the University of Cambridge
WHAT: Built a $10 light sensor that works like a big lab machine
WHERE: In England
WHY: To make accurate testing small and cheap enough for everyday devices

Why it matters

Small, cheap sensors could one day help people check their health with something as simple as a watch.

Test what you remember

1. About how much does the new sensor cost?

- (A) $1
- (B) $1 million
- (C) $500
- (D) $10

2. Where was the sensor built?

- (A) University of Cambridge
- (B) In a bakery
- (C) On the Moon
- (D) At a zoo

3. What does a spectrometer study?

- (A) Smell
- (B) Light
- (C) Taste
- (D) Sound

4. Which item did the sensor identify correctly?

- (A) Music
- (B) Coffee
- (C) Clouds
- (D) Rainbows

5. What might it help people check one day without needles?

- (A) Shoe size
- (B) Hair color
- (C) Blood sugar
- (D) Eye color

6. What do scientists say is still needed?

- (A) A new name
- (B) Nothing
- (C) A bigger box
- (D) More testing

Answer key (try the quiz first!): 1: D 2: A 3: B 4: B 5: C 6: D

Questions to think about

MY THOUGHTS (at least 20 words):